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Optimized Machine Learning-Based Survival Prediction in HCC: Addressing Data Imbalance and Feature Selection

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***Abstract*— Hepatocellular Carcinoma (HCC) is still one of the most lethal liver tumors worldwide, and also a challenging problem for early survival prediction due to the heterogeneous and class-imbalanced data among patients. This work introduces an ensemble machine learning model to predict one-year survival in HCC using a real-world dataset comprising 165 patient samples and 49 distinct clinical features. Preprocessing was performed in a rigorous manner: missing value imputation, normalization, and systematic feature selection utilizing the SelectKBest method and Chi-square test. SMOTEENN (Synthetic Over Sampling and Edited Nearest Neighbors) was applied to resolve the imbalance. The proposed experimental results demonstrate that it is indeed beneficial to choose an optimal subset of features, and obtain an impressive test performance of 97.06%. Through combining tuned XGBoost, LightGBM, and CatBoost models in a stacked ensemble and Logistic Regression as the meta learner, the proposed approach exhibits enhanced robustness and generalization, providing decision support for HCC treatment planning with better accuracy and individualization.**

Keywords*— Hepatocellular Carcinoma, SMOTEENN, SelectKBest, Chi-square test, Ensemble, Feature Selection.

I. INTRODUCTION

The most lethal primary liver cancer worldwide is hepatocellular carcinoma (HCC). D. Sia et al. [1] stressed the importance of knowledge about the cellular origin and molecular subtypes of HCC, which have major implications for accurate diagnosis and treatment. Their results emphasized the heterogeneity of HCC tumorigenesis driven by various gene and epigenetic changes. These variations lead to molecular subtypes with different prognoses and responses to treatment. This emphasizes the need for personalized medicine specific to these subtypes.

P. Vande Lune et al. [2] gave an extensive review of the diagnosis, therapy, and post-therapy imaging of HCC, with particular emphasis on TACE. They observed that although the predictability of survival has improved, due to biological variability and the absence of highly predictive models, predicting survival remains difficult. This represents an unmet need for computational tools that integrate imaging with clinical data through AI and machine learning, enabling improved accuracy and personalized treatment planning.

In line with the computational models, M.R. Haque et al. [3] studied the use of Random Forest (RF) and Artificial Neural Networks (ANN) and compared these techniques for classifying liver diseases. Their work proved that ensemble methods can efficiently handle nonlinear relationships and noise, which is crucial and suitable for complicated diseases such as HCC. They emphasized that model performance is greatly dependent on preprocessing, the choice of features, and dealing with class imbalance, also corroborating that ensemble learning increases the predictive performance.

Similarly, S. Hashem et al. [4] showed that machine learning models with appropriate regularization and sophisticated feature pre-processing can notably enhance diagnostic performance for HCC, particularly for HCV-related liver disease patients. It should be noted that they focused on (imputation of) missing values, feature scaling, as well as the imbalanced classes problem, in order to improve the sensitivity score. Against this

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backdrop, the proposed work presents a stateof-the-art solution as follows:

- The proposed ensemble architecture is a combination of XGBoost, LightGBM, and CatBoost in a stacking fashion, using Logistic Regression as a meta-learner.
- KNN imputation is used, Min-Max scaling, SelectKBest with Chi-square for feature selection, and SMOTEENN for class balancing.
- The proposed pipeline achieved an accuracy of 97.06%

This illustrates that a combination of robust preprocessing, advanced feature engineering, and ensemble learning could facilitate the improvement of the prediction of HCC survival and personalized patient care.

II. Related Work

In this section, the related works on the prediction of hepatocellular carcinoma with different machine learning and other techniques are discussed. L. Ali et al. [5] proposed the LDA-based hybrid classifier based on the combination of LDA for dimensionality reduction, GA for SVM parameter optimization, and SVM for classification and observed that careful feature selection in the tuning process is of great help in performance in the most challenging, small, nonhomogeneous HCC datasets. W. Ksi ˛azek et al. [6] proposed an early ˙ detection pipeline which compares decision trees, random forests, and SVMs, emphasizing that ensembles can handle noise and nonlinearity better, and emphasizing the importance of balanced datasets and cross-validation ideas underpinning the stacked generalization approach. L. Guo et al. [7] applied RF to unify clinical and genomic data for prognosis, showing ensembles are useful for their interpretation as well as the prediction, consistent with the SelectKBest method.

R. Sawhney et al. [8] demonstrated that the metaheuristic Firefly Algorithm based selection of features enhanced generalization and eliminated irrelevant ones, thereby enabling robust selection as well, and improved prediction accuracy to 83.5%, in line with SMOTEENN and robust selection. M. Hattab et al. [9] employed SMOTE with instance selection for early HCC detection, thus validating the requirement of an oversampling for fair training of imbalanced datasets, which in turn supports the effectiveness of the hybrid SMOTEENN approach. G. Choi et al. [10] proposed a CDSS that integrated multiple types of data, suggesting that predictive accuracy and interpretability are equally important for clinical trust as we offer through stacked gradient boosting models with a Logistic Regression meta-learner.

X. Jiang et al. [11] demonstrated the effectiveness of LightGBM to predict HCC. F. Al-Islam et al. [12] supported the advantage of SMOTE with strong classifiers such as XGBoost, with 87% accuracy. Wang et al. [13] demonstrated that integrating serological and demographic features with ensemble learning algorithms such as Random Forest, LightGBM, XGBoost, and CatBoost resulted in a considerable enhancement in the early diagnosis of Primary Hepatocellular Carcinoma (PHC) with an accuracy of 96.62%. Through the selection of an optimal subset of features, they demonstrated how ensemble models can learn subtle patterns that have been overlooked by traditional methods, to facilitate more accurate, clinically actionable screening for PHC. Altogether, these studies are consistent with the proposed methodology, combining oversampling, feature selection, and stacked ensemble (XGBoost + LightGBM + CatBoost) with interpretability to deliver robust, generalizable, and trusted HCC survival predictions tailored for clinical decision-making.

III. Methodology

The proposed system architecture for HCC survival prediction is shown in Fig. 1, which consists of a traditional ML pipeline including two workflows and an advanced ensemble pipeline. The data is collected, pre-processed, divided, and balanced. The baseline was partially implemented with mean imputation, label encoding, scaling, SMOTE, and traditional classifiers, including Logistic Regression, SVM, and XGBoost. On the other hand, the ensemble pipeline uses KNN imputation, SMOTEENN for better balancing, and SelectKBest with Chi 2 for selecting the subset of features. Then, the metalearner model is constructed, Logistic Regression, to improve robustness and understandability. Both pipelines are compared, and the ensemble methodology is shown to be more proficient in dealing with intricate patterns and data imbalance.

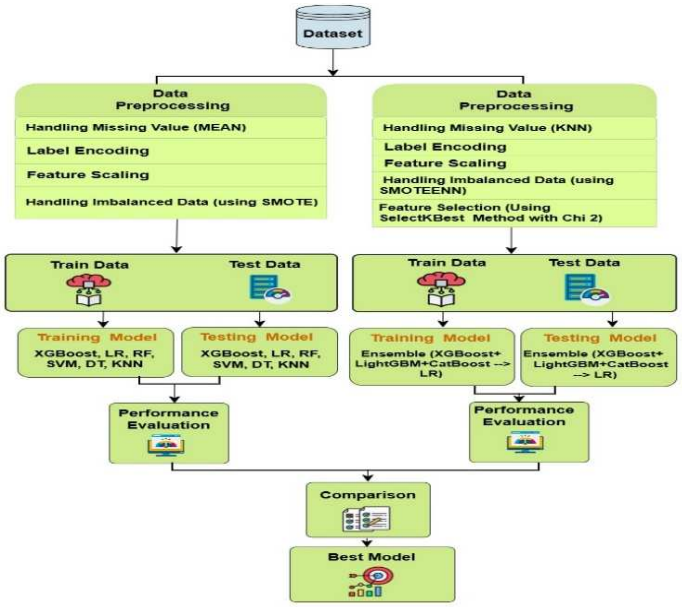


Fig. 1: Proposed computational framework for HCC survival prediction

A. Description of the Dataset

The dataset used for this study refers to patients with HCC at a University Hospital in Portugal. It has been used in multiple studies for survival analysis and clinical prognosis, and is available from the UCI Machine Learning Repository [14]. Overall, the dataset consists of 165 cases and 49 features defined according to the EASL-EORTC (European Association for the Study of the Liver – European Organization for Research and Treatment of Cancer). Distribution of variable types in the HCC dataset is shown in Fig 2. The dataset is heterogeneous and

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consists of continuous and discrete numerical features and nominal and ordinal categorical features— 23 numerical and 26 categorical attributes. These factors are diverse and include patient characteristics such as age and sex, risk factors (such as alcohol consumption, viral infections, and diabetes), clinical parameters (AFP, bilirubin, hemoglobin), and imaging information related to diagnosis. One significant feature about this dataset is the missing data, which amounts to 10.22% of total entries. This demonstrates that the data is real-world and that it needs preprocessing. The target variable, named "Class," represents the survival status after one year and it is coded with Class = 1 for 'Survived' and Class = 0 for 'Died within one year'. The dataset also has a class imbalance; from these 165 cases, 102 (61.8%) are categorized as "survived" and 63 (38.2%) as "died". This imbalance further complicates the problem for machine learning models, which, during training, will naturally skew toward the majority class, which needs to be balanced by applying appropriate techniques.

B. Data Preprocessing

The dataset used in this study exhibited missing values, which are a common challenge in clinical datasets. Missing data can significantly affect the statistical power of the analysis, bias parameter estimation, and reduce the generalizability of the models developed from such data. In this dataset, missing entries were represented by the symbol "?". These placeholders were first converted into NaN (Not-a-Number) values for compatibility with data handling libraries. To address the missingness, the mean value is employed, a widely adopted method that replaces missing numerical values with the average of available entries in the respective feature. Later, KNN imputation is used for better results. While imputation provides a simple and effective solution, it can still introduce bias if the data are not missing completely at random (MCAR) [15].

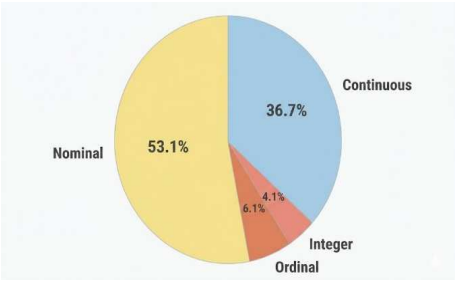


Fig. 2: Distribution of variable types in the HCC dataset.

Additionally, because many features in the dataset were categorical and machine learning algorithms usually work better with numbers, label encoding was used to change the categorical variables into numbers. Label encoding ensures that each unique category is mapped to an integer, making the data interpretable by algorithms that expect numerical input. In addition, feature scaling was performed to bring all numerical values into a consistent range. Specifically, MinMax normalization was applied, which transforms feature values to fall within the range [0, 1], using the formula: The principle for applying normalization is in (1).

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

Here, X_{max} and X are the highest and the lowest esteem of the factor, correspondingly. This type of normalization is especially important when different features have varying units and scales, as it ensures that all inputs contribute proportionately to the learning algorithm [16].

C. *Imbalanced Data Handling by* SMOTE/ *SMOTEENN*

In imbalanced class distributions, one class has tens or hundreds of samples, while the other may have only a few samples, which challenges the equal-class assumption of many algorithms. As a result, the accuracy of predictions degrades, in particular with respect to the small (but often most important) minority class [17].

The SMOTE (Synthetic Minority Over-sampling Technique) algorithms oversamples the minority class by interpolating between every minority point and its k nearest neighbours to generate realistic synthetic samples rather than pure duplicates [18], [19].

In this paper, the training instances are sampled using SMOTEENN, oversampling the minority class with SMOTE and then balancing with Edited Nearest Neighbours (ENN). This helped balance the class distribution as Table I indicates. But SMOTE can introduce noise by itself. To address this, in the case of ENN, it is used to clean the dataset by removing the unclear or noisy instances from both the minority and majority classes. Further fine-tuned the ENN component in the final model by n_neighbors=2 in EditedNearestNeighbours, which removes borderline and ambiguous samples more aggressively during data cleaning, it can more aggressively refine the dataset, so that the decision boundary is clearer between survival and death classes.

SMOTE and ENN cooperate to refine the dataset on survival vs. death and make the balance, which is conducive to the good performance of the model in classifying both survival and death. This approach has the dual benefit of balancing training data and making the model more sensitive and accurate, particularly for the minority class. This results in a more credible and impartial inference, which is crucial in clinical decision support systems.

D. *Machine Learning for Classification*

After all the pre-processing, several machine learning methods (Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), and XGBoost) were investigated on the dataset for survival status prediction of HCC patients. These classifiers are selected because of their success in classification and application to the biomedical field. Each algorithm has its own strong points: Logistic Regression is a method that is easy to interpret and to understand, it is very efficient to train; it should be the first algorithm an analyst should pick; Decision Trees / Random Forests are good to handle non-linear patterns and they provide an implicit feature selection; SVMs are useful

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in high dimensional spaces and are still one of the most accurate algorithms; KNN is often considered lazy, although it is a great algorithm. XGBoost, as a gradient boosting-based algorithm, is famous for its high accuracy and fast scalability. The mean imputer is used in data pre-processing, SMOTE as a data balancing method, and no feature is selected. Moreover, to the above individual models, an ensemble-based approach is constructed. This model aggregates the predictions from several models, i.e., XGBoost, LightGBM, and CatBoost, with a meta-classifier (Logistic Regression) stacked on top. Ensemble methods are also appealing because they leverage the power of multiple models and can yield superior generalization and performance on unseen data.

E. *Ensemble ML-Based Approach*

To improve prediction performance and address the limitations based on single classifiers, an Ensemble Machine Learning (ML) [20, 21, 22] Model was developed by aggregating multiple high-performance algorithms. The effectiveness of the ensemble arises from the diversity and complementary strengths of the constituent models which in noisy and complex data, such as medical datasets, a combination of multiple diverse models usually outperforms any single model.

In the ensemble model, XGBoost, LightGBM, and CatBoost, three strong gradient boosting algorithms are used. These three models have been selected because they performed well in previous classification problems, are able to manage missing values and categorical values, and are known to be immune to the presence of the class imbalance and the interaction influence between the features.

All base models (i.e., XGBoost, LightGBM, and CatBoost) were trained separately using the balanced dataset balanced by SMOTEENN. The final output predictions (probabilities) generated by the base learners are then fed into a metaclassifier (implemented via the Logistic Regression). This architecture creates a stacked ensemble model, with the base models being the level-0 learners while the second model is the meta-classifier or level-1 learner. The workflow of the proposed ensemble model is shown in Fig 3.

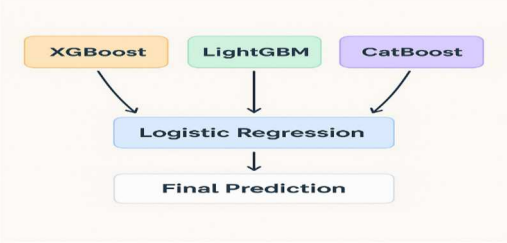


Fig. 3: Proposed ensemble model workflow.

The meta-classifier was trained to maximize the final prediction using the aggregated information from the base models. The simplicity and interpretability of the Logistic Regression model are the reasons it was used as the meta-learner and was shown to be effective and not compromising much the performance of prediction. The meta-classifier learns from the output of the data level and is in a better position to take a final decision.

IV. RESULTS AND DISCUSSIONS

In this study, we systematically evaluated the predictive power of both conventional machine learning classifiers and our proposed ensemble framework for Hepatocellular Carcinoma (HCC) 1-year survival prediction. To establish a fair baseline, six well-known classifiers — Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), and XGBoost were first tested individually. All baseline models used the same pre-processed dataset, where missing values were handled with mean imputation and balanced using SMOTE to address the class imbalance challenge, following standard practice in medical ML studies. The balanced dataset increased the training instances from 132 to 166, as shown in Table I, ensuring each class (survives/dies) was well represented. The classifiers were trained with an 80:20 split for training and testing to assess their ability to generalize on unseen data.

TABLE I. SMOTE IMPACT ON DATASET

Before Applying SMOTE		After Applying SMOTE	
No. of Training Data	No. of Test Data	No. of Training Data	No. of Test Data
132	33	166	38

We measured our implemented system performance based on accuracy, precision, Specificity, F-measure, and Sensitivity using the following formulas in (2), (3), (4), (5), and (6), respectively.

$$Accuracy = \frac{TP + TN}{TP+FP+FN+TN} \quad (2)$$

$$Precision = \frac{TP}{TP+FP} \quad (3)$$

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (4)$$

$$\text{F-measure} = 2 * \frac{(Precision*Sensittivity)}{(Precision+Sensittivity)} \quad (5)$$

$$Sensitivity = \frac{TP}{TP+FN} \quad (6)$$

TABLE II. PERFORMANCE COMPARISON OF SINGLE CLASSIFIER MODEL

Model	F-Measure	Specificity	Sensitivity	Precision	Accuracy
XGBoost	**93.02%**	**76.92%**	**100%**	**86.96%**	**87.88%**
LR	84.21%	84.62%	80%	88.89%	81.82%
RF	88.37%	69.23%	95%	82.61%	84.85%
SVM	74.42%	46.15%	80%	69.57%	66.67%
DT	68.29%	46.15%	70%	66.67%	60.61%
KNN	77.27%	46.15%	85%	70.83%	69.7%

As seen, the XGBoost Model performed better than all individual classifiers, which achieved 87.88% accuracy, emerging as the best single model. Random Forest and Logistic Regression followed with 84.85% and 81.82% accuracy, respectively. KNN achieved 69.70%, SVM resulted in 66.67%, and Decision Tree performed the lowest at 60.61%. These variations illustrate how advanced gradient boosting methods (like XGBoost) handle complex patterns and feature

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interactions better than simpler models. Table II shows the performance comparison of different ML classifiers

Then we implement an ensemble framework for Hepatocellular Carcinoma (HCC) 1-year survival prediction. This model used the pre-processed dataset, where missing values were handled with KNN imputation and balanced using SMOTEENN to address the class imbalance challenge. The balanced dataset increased the training instances from 132 to 166, like Table I, ensuring each class (survives/dies) was well represented. The classifiers were trained with an 80:20 split for training and testing to assess their ability to generalize on unseen data. For getting a better predictive capability of the ensemble machine learning model, a systematic feature selection experiment was conducted using the SelectKBest method with the Chi-Square. The main goal of this step was to identify an optimal subset of features that contribute most significantly to predicting survival in hepatocellular carcinoma (HCC) patients, while reducing noise and avoiding overfitting.

In this experiment, we iteratively evaluated the model's performance for various feature counts. For each feature count, we compare the classification accuracy alongside the specific feature names selected and found that the feature count of 19 features emerged as a balanced, robust option: it achieved a solid accuracy of 97.06% indicating strong generalization across folds by using the SelectKBest method with the Chi-Square. This suggests that the selected 19 features (shown in Fig. 4) provide sufficient discriminatory power while minimizing redundant or irrelevant variables.

The Ensemble Model, taking nineteen features, performed better than all single classifiers. It was the best in terms of Accuracy (97.06%), F1-score (96%), Precision (92.31%), Sensitivity (Recall) (100%), and Specificity (95.45%). These findings illustrate in Fig 5 shows that the proposed ensemble model is extremely powerful. The high F1-score indicates a strong balance between Precision and Recall that is critical for a clinical application. In other words, the perfect Recall (100%) indicates the model never misses a true non-survivor, which implies that it removes false negatives, discovering all non-survivors. Meanwhile, the high Precision (92.31%) also indicates that the number of false positives is low. This trade-off ensures that the ensemble can provide a reliable prediction for the at-risk patients without compromising on the overall predictive performance, which is important for reliable and actionable clinical decision support. The confusion matrix of proposed ensemble model is shown in Fig. 6.

Among the selected features, several clinical variables demonstrated strong predictive importance. The top ten most influential features identified by the Chi-Square scoring method include Alpha-Fetoprotein (AFP), Total Bilirubin, Hemoglobin, Platelet Count, Albumin, Tumor Size, Ascites, International Normalized Ratio (INR), Age, and Portal Vein Thrombosis. These variables are clinically recognized indicators associated with liver function, tumor progression, and overall survival outcomes in hepatocellular carcinoma patients.

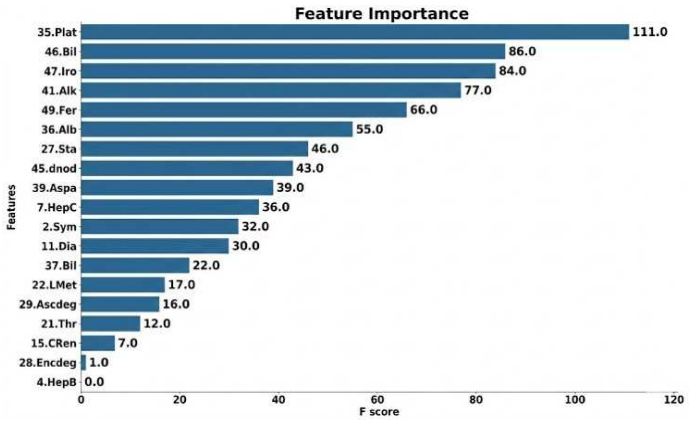


Fig 4: Feature Importance score of selected features

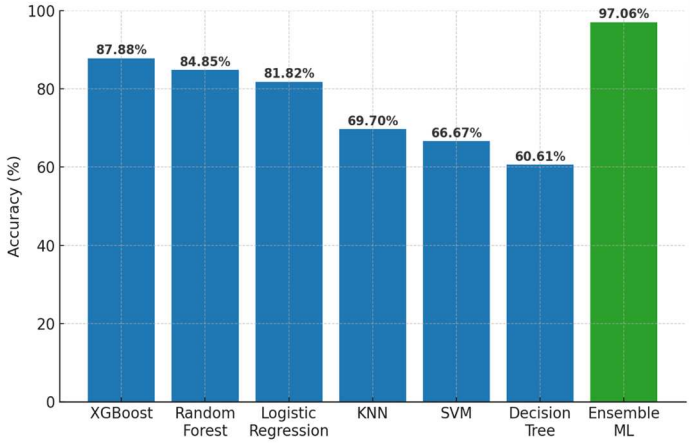


Fig 5: Accuracy comparison between different single classifier models and the Proposed Ensemble Model

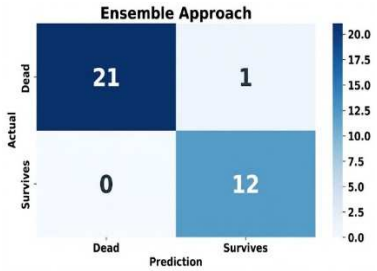


Fig 6: Confusion Matrix of Ensemble ML Based Approach

Our results align with and surpass comparable research. In table III, it is observed that Sawhney et al. [8] combined Firefly and RF for 83.5% accuracy, while Hattab et al. [9] achieved 84.90% with SVM on a similar dataset. Ahmed et al. [11] achieved 89% with Ensemble LightGBM. Ferdib-Al-Islam et al. [12] reached 87% with XGBoost plus SMOTE. Wang et al. [13] reached 96.62% accuracy using an ensemble Model. In contrast, our hybrid Ensemble ML-Based Approach with SMOTEENN and SelectKBest with Chi-Square pushed accuracy up to 97.06% - a significant gain in both efficiency and reliability.

TABLE III. PROPOSED MODEL'S COMPARISON

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PAPER	MODEL	ACCURACY
SAWHNEY ET AL. [8]	FIREFLY + RANDOM FOREST	83.5%
HATTAB ET AL. [9]	LOGISTIC REGRESSION	81.08%
	SVM	84.90%
FERDIB-AL-ISLAM ET AL. [12]	DECISION TREE	76%
	XGBOOST	87%
WANG ET AL. [13]	RANDOM FOREST +XGBOOST + LIGHTGBM + CATBOOST	96.62%
PROPOSED ENSEMBLE ML BASED MODEL	**LOGISTIC REGRESSION+XGBOOST + LIGHTGBM + CATBOOST**	**97.06%**

V. CONCLUSIONS

In this paper, we focused on the urgent and significant predictions of survival for HCC patients based on an ensemble approach of machine learning. Based on a real-world dataset with 165 subjects with 49 clinical features, we conducted robust preprocessing, advanced feature selection, and the hybrid sampling approaches with SMOTEENN to cope with the class imbalance issue reasonably. A systematic investigation demonstrated that meticulous feature selection has a substantial influence on model performance, accuracy of 97.06% was obtained using the top nineteen features. Our method achieves better generalization performance than constituting single models running tuned XGBoost, LightGBM, and CatBoost classifiers with a stacking ensemble. This study further demonstrates that integration of state-of-the-art oversampling, feature selection, and ensemble learning can significantly enhance the prediction accuracy of HCC survival and is favorable in clinical decision-making at the early stage in a patient-specific manner.

References

[1] D. Sia, A. Villanueva, S. Friedman, and J. Llovet, "Liver Cancer Cell of Origin, Molecular Class, and Effects on Patient Prognosis," Gastroenterology, vol. 152, no. 4, pp. 745–761, 2017.

[2] P. Vande Lune, A. Abdel Aal, S. Klimkowski, J. Zarzour, and A. Gunn, "Hepatocellular Carcinoma: Diagnosis, Treatment Algorithms, and Imaging Appearance after Transarterial Chemoembolization," J. Clin. Transl. Hepatol., vol. 6, no. 2, pp. 114, 2017.

[3] M. R. Haque, M. M. Islam, H. Iqbal, M. S. Reza, and M. K. Hasan, "Performance Evaluation of Random Forests and Artificial Neural Networks for the Classification of Liver Disorder," in Proc. Int. Conf. Computer, Communication, Chemical, Material and Electronic Engineering (IC4ME2), 2018, pp. 1–5.

[4] S. Hashem et al., "Machine Learning Prediction Models for Diagnosing Hepatocellular Carcinoma with HCV-related Chronic Liver Disease," Comput. Methods Programs Biomed., vol. 196, p. 105551, 2020.

[5] L. Ali, I. Wajahat, N. A. Golilarz, F. Keshtkar, and S. A. C. Bukhari, "LDA-GA-SVM: Improved hepatocellular carcinoma prediction through dimensionality reduction and genetically optimized support vector machine," Neural Comput. Appl., pp. 1–10, 2020.

[6] W. Książek, M. Abdar, U. Acharya, and P. Plawiak, "A novel machine learning approach for early detection of hepatocellular carcinoma patients," Cogn. Syst. Res., vol. 54, pp. 116–127, 2019.

[7] L. Guo et al., "Random-forest algorithm based biomarkers in predicting prognosis in the patients with hepatocellular carcinoma," Cancer Cell Int., vol. 20, no. 1, 2020.

[8] R. Sawhney, P. Mathur, and R. Shankar, "A Firefly Algorithm Based Wrapper-Penalty Feature Selection Method for Cancer Diagnosis," 2020.

[9] M. Hattab, A. Maalel, and H. Ghezala, "Towards an Oversampling Method to Improve Hepatocellular Carcinoma Early Prediction," Digit. Health Focus Predictive, Preventive and Personalised Med..

[10] G. Choi et al., "Development of machine learning-based clinical decision support system for hepatocellular carcinoma," Sci. Rep., vol. 10, no. 1, 2020.

[11] X. Jiang, R. Zhou, F. Jiang, Y. Yan, Z. Zhang, and J. Wang, "Construction of diagnostic models for the progression of hepatocellular carcinoma using machine learning," Front. Oncol., vol. 14, Art. no. 1401496, 2024.

[12] F. Al-Islam, L. Akter, and M. M. Islam, "Hepatocellular Carcinoma Patients' Survival Prediction Using Oversampling and Machine Learning Techniques," in Proc. 2nd Int. Conf. Robotics, Electr. Signal Process. Tech. (ICREST), 2021, pp. 445–450, doi: 10.1109/ICREST51555.2021.9331108.

[13] X. Wang, Y. Zhang, L. Li, *et al*., "Ensemble learning enhances the precision of preliminary detection of primary hepatocellular carcinoma based on serological and demographic indices," *Front. Oncol.*, vol. 14, Art. no. 38952558, 2024.

[14] "UCI Machine Learning Repository: HCC Survival Data Set," Archive.ics.uci.edu, 2020. [Online]. Available: http://archive.ics.uci.edu/ml/datasets/HCC+Survival. [Accessed: 10-March-2025].

[15] H. Kang, "The prevention and handling of the missing data," Korean J. Anesthesiol., vol. 64, no. 5, pp. 402–406, 2013.

[16] S. Patro and K. Sahu, "Normalization: A Preprocessing Stage," Int. Adv. Res. J. Sci. Eng. Technol., vol. 2, no. 3, pp. 20–22, 2015.

[17] M. Rahman and D. Davis, "Addressing the Class Imbalance Problem in Medical Datasets," Int. J. Mach. Learn. Comput., vol. 3, no. 2, pp. 224–228, 2013.

[18] K. Wang, B. Makond, K. Chen, and K. Wang, "A hybrid classifier combining SMOTE with PSO to estimate 5-year survivability of breast cancer patients," Appl. Soft Comput., vol. 20, pp. 15–24, 2014.

[19] S. F. Abdoh, M. Abo Rizka, and F. A. Maghraby, "Cervical Cancer Diagnosis Using Random Forest Classifier With SMOTE and Feature Reduction Techniques," IEEE Access, vol. 6, pp. 59475–59485, 2018.

[20] Z.-H. Zhou, "Ensemble learning," in Encyclopedia of Biometrics, Springer, pp. 411–416, 2009.

[21] L. Rokach, "Ensemble-based classifiers," Artif. Intell. Rev., vol. 33, no. 1–2, pp. 1–39, 2010.

[22] N. Prokhorenkova et al., "CatBoost: unbiased boosting with categorical features," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 31, pp. 6638–6648, 2018.

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